

**GREAT LAKES UNIVERSITY
UNIVERSITY OF KISUMU
COURSE OUTLINE**

SCHOOL: Community Health & Development
DEPARTMENT:
CAMPUS/SEMESTER: Kisumu Campus/January-April 2026

COURSE CODE/TITLE: : BIostatistics

COURSE LECTURER: Christine Akinyi Odieny **SIGNATURE:**
CONTACT: odienychristine42@outlook.com

COURSE DESCRIPTION: Basic statistical measurements of central tendency and dispersion, means, median, hypothesis testing, estimation, confidence intervals, correlation and regression analysis; Probability and Inference. Theoretical Distributions: The principles, assumptions, on Normal, Binomial and Poisson distributions. Sampling techniques: Determination of sample size: Sample size estimation for various types of studies; case-control, evaluative, cross-sectional, prospective, clinical trials studies. Hypothesis Testing: Null Hypothesis, Alternative Hypothesis: Level of confidence; Type I and Type II errors; Power of the Test; Confidence intervals; Testing the difference between two sample means and proportions; Chi-Square and Student t-test, McNamar's test. Regression: Multiple regressions, logistic regression, fitting of model to data, confidence intervals for regression line, interpretation of regression outputs, meaning of R^2 Pearson coefficient. Analysis of variance: (ANOVA) Two-by-two tables, multi tables, assumptions, analysis, and interpretation of ANOVA tests.

ELOs: By the end of the course the learner will:

- i) Enhance understanding of statistical measures and inference through practical application and analysis.
- ii) Acquire a comprehensive knowledge of theoretical distributions through the study and application of their properties, characteristics, and statistical inference techniques.
- iii) Develop proficiency in sampling techniques and the determination of sample size, demonstrated through the

application of appropriate sampling methods and accurate sample size calculations in practical research scenarios.

- iv) Attain competence in hypothesis testing, regression analysis, and analysis of variance (ANOVA) through the successful application of these statistical methods in data analysis, interpretation of results, and drawing meaningful conclusions based on statistical inference.

Week	Topic	ELOs	Sub-Topic
1	Introduction to Biostatistics and Data Description	1. Describe the basic principles of biostatistics and its importance in healthcare research. 2. Apply appropriate measures of central tendency and dispersion to summarize and interpret biomedical data. 3. Describe different types of data and variables encountered in biostatistical analysis.	• Types of data and variables • Descriptive statistics: measures of central tendency and dispersion • Calculation and interpretation of means, medians, and measures of variability
2	Probability and Theoretical Distributions	1. Apply probability concepts and rules to calculate probabilities in the context of healthcare data. 2. Understand the principles and assumptions underlying the Normal, Binomial, and Poisson distributions in biostatistics. 3. Apply the properties of these distributions to model and analyze healthcare data.	• Basic concepts of probability theory • Probability rules and calculations • Probability distributions: Binomial, Poisson, Normal, t-, F- and Chi-square
3	Sampling and Sampling Techniques	1. Differentiate between Probability and Non-probability sampling approaches, understand their advantages and disadvantages, and apply appropriate sampling techniques based on the research context. 2. Explain the concept of sampling distributions and their role in statistical inference. 3. Apply the central limit theorem to estimate population parameters and make inferences about healthcare datasets.	• Probability and Non-Probability Sampling Techniques • Sampling distributions and the central limit theorem
4	Confidence Intervals	1. Construct and interpret confidence intervals for means, proportions, and other parameters in biostatistics. 2. Determine appropriate sample sizes for different types of studies in healthcare research. 3. Evaluate the precision and reliability of estimates using confidence intervals in the context of clinical trials and epidemiological studies.	• Construction and interpretation of confidence intervals • Confidence intervals for means and proportions • Sample size determination for different types of studies ○ Case-control, evaluative, cross-sectional, prospective, and clinical trials studies
5	Hypothesis Testing	1. Formulate null and alternative hypotheses for research questions in biostatistics. 2. Calculate and interpret p-values and make decisions based on hypothesis tests in the context of healthcare studies. 3. Understand the concepts of Type I and Type II errors and their implications in hypothesis testing in biostatistics.	• Null and Alternative hypotheses • Type I and Type II errors • Significance level, p-value, and critical regions • One-sample and two-sample hypothesis tests ○ Student's t-test and its applications • Chi-Square test for independence
6	ANOVA	1. Understand the concept and assumptions of ANOVA and its applications in healthcare research.	• Introduction to ANOVA • One-way ANOVA: assumptions, calculations, and interpretation.

		- Perform one-way ANOVA to compare means of multiple groups in biomedical studies. - Interpret ANOVA results, including F-statistics and p-values, and conduct post-hoc tests for multiple comparisons.	- Two-way ANOVA and multi-way ANOVA. - Post-hoc tests and multiple comparisons
7	Non-Parametric Tests	- Identify situations where non-parametric tests are applicable in biostatistical analysis. - Apply non-parametric tests such as the Wilcoxon signed-rank test, Mann-Whitney U test, and Kruskal-Wallis test to healthcare datasets. - Interpret non-parametric test results and understand their advantages and limitations compared to parametric tests.	- Introduction to non-parametric tests - Wilcoxon signed-rank test - Mann-Whitney U test - Kruskal-Wallis test - McNemar test
8	Correlation and Regression Analysis	- Analyze and interpret the strength and direction of relationships between variables in healthcare datasets using correlation coefficients. - Apply simple linear regression analysis to model and predict health outcomes based on explanatory variables. - Interpret regression outputs, including coefficients, p-values, and R-squared, in the context of biomedical research.	- Understanding the relationship between variables - Calculation and interpretation of correlation coefficients - Simple linear regression analysis - Fitting a regression model to data - Confidence intervals and hypothesis testing for regression coefficients - Introduction to logistic regression
9	Advanced Topics in Statistical Inference	- Understand the concept of statistical power and its importance in designing and interpreting healthcare studies. - Determine sample sizes for hypothesis testing and power calculations in biostatistics. - Apply meta-analysis techniques to combine and analyze results from multiple studies in healthcare research.	- Power of the test - Sample size determination for hypothesis testing - Non-inferiority and equivalence testing - Meta-analysis: combining results from multiple studies
10	Review and Applications	- Review and consolidate key concepts and techniques covered throughout the course. - Apply statistical methods and reasoning to real-world scenarios in biostatistics. - Demonstrate critical thinking and ethical considerations in statistical analysis and data interpretation in the healthcare field.	- Recap of key concepts and techniques covered in the course - Application of statistical methods to real-world scenarios - Ethical considerations in statistical analysis

ASSESSMENT STRATEGY: Each week will consist of lectures, practical examples, and problem-solving sessions.
Assignments and quizzes will be given to reinforce understanding and assess learning progress.
End of Semester Examination

REFERENCES:

- Bernard Rosner (2016) Fundamentals of Biostatistics 8th Edition. CENGAGE Learning. ISBN: 978-1-305-26892-0
- Chang, Mark. (2013). Modern Issues and Methods in Biostatistics (Statistics for Biology and Health).

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example, one microarray experiment may involve several thousand genes. If we set the inference error rate at 5%, we could expect to see hundreds of invalid inferences in one experiment. To guard against such inflated error rate, we often need to control the experiment-wise error (i.e., the overall chance of making an incorrect inference) rather than the error associated with a specific inference. A general class of methods for such adjustments are often referred to as simultaneous inference.

One of the popular simultaneous inference procedures is called the Bonferroni method. The basic idea of Bonferroni's approach is to reduce the individual inference error rate to a lower level so that the experiment-wise error rate can be controlled at the nominal level of $\alpha 100\%$. Specifically, if there are k inferences, we construct the $(1 - \alpha)100\%$ confidence interval using $t_{\alpha/(2k)}$ instead of $t_{\alpha/2}$.

Bonferroni's adjustment is based on the well-known Bonferroni inequality, a probability *inequality*. So the method is not exact, (i.e., the resulting experiment-wise error rate would be *no more than* α). At times, the Bonferroni adjustment may result in an experiment-wise error rate much less than the stated level. Therefore, it is often regarded as one of the more conservative multiplicity adjustment procedures.

Using the Bonferroni procedure, we can calculate the mean BNP levels for NYHA classes II and III patients as (255.6, 568.4) and (315.4, 1146.6), respectively. Note that these intervals are considerably wider than the unadjusted ones, suggesting the loss of precision as a result of the increased error rate. It should be pointed out that the Bonferroni adjustment may be too conservative (i.e., resulting intervals are too wide) for many applications. To alleviate, many alternative adjustment methods have been proposed (see **Chapter 7** for a more in-depth discussion on the subject).

2.2.5. Confidence Interval for the Variance

Occasionally, our research requires inference on population variances. In **Section 2.1**, we have stated that the point estimate of the population variance σ^2 is its sample counterpart s^2 (i.e., $\hat{\sigma}^2 = s^2$). This section describes the construction of a confidence interval estimate of σ^2 using the sampling distribution of s^2 . Specifically, let $X_1, \dots, X_n$ be a random sample from a normally distributed population with mean μ and variance σ^2 . The quantity $(n - 1)s^2/\sigma^2$ follows a chi-square distribution with $n - 1$ degrees of freedom, that is,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{(n-1)}^2.$$

We therefore have

$$P\left(\chi_{1-\alpha/2}^2 \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi_{\alpha/2}^2\right) = 1 - \alpha,$$

where $\chi_{1-\alpha/2}$, and $\chi_{\alpha/2}$ respectively correspond with the $(1 - \alpha/2)$ th and $(\alpha/2)$ th quantiles in a chi-square distribution with $n - 1$ degrees of freedom. From this, we obtain the $100(1 - \alpha)\%$ confidence interval estimate of variance σ^2 as

$$\frac{(n-1)s^2}{\chi_{\alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}.$$

Example 3

To estimate the variation of a particular model of bone density scanner, the manufacturer of the scanners randomly selected four of its machines for testing. In measuring a standard of known density, the four selected scanners produced the following readings: 4.1, 4.0, 3.9, 3.9. Give a 95% confidence estimate of the variance of this model of bone scanner when the scanners are used to measure the standard. We first calculated the sample variance $s^2 = 0.0092$. The 95% confidence estimate of the variance of this model of bone scanner is then

$$\begin{aligned} \frac{(n-1)s^2}{\chi_{\alpha/2}^2} &\leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2} \\ \frac{(4-1) \times 0.0092}{9.3484} &\leq \sigma^2 \leq \frac{(4-1) \times 0.0092}{0.2058} \\ 0.0030 &\leq \sigma^2 \leq 0.1341, \end{aligned}$$

where 0.2058 and 9.3484 are the 2.5th and 97.5th percentiles, respectively, of the chi-square distribution with 3 degrees of freedom. Based on the above calculation, we can claim with 95% confidence that the variance of the measuring instruments is between 0.003 and 0.1341.

2.2.6. One-Sided Confidence Intervals

The discussion of confidence intervals so far has been of two-sided intervals. That is, we provide lower and upper limits on the parameter of interest. We can also define one-sided confidence intervals that consist only of a lower confidence limit or only an upper confidence limit. A lower one-sided $100(1 - \alpha)\%$ confidence interval estimate of θ satisfies the following criterion

$$P(\hat{\theta}_1 \leq \theta) = 1 - \alpha.$$

An upper one-sided $100(1 - \alpha)\%$ confidence interval estimate of θ satisfies the following criterion

HYPOTHESIS TESTING

BASIC PRINCIPLES OF HYPOTHESIS TESTING

Definition of Hypothesis Testing

- Hypothesis testing is a statistical method used to make inferences or draw conclusions about a population based on a sample of data.
 - The process involves formulating two competing hypotheses, the null hypothesis (H_0) and the alternative hypothesis (H_1 or H_A), and assessing the evidence from the sample to determine which hypothesis is more supported by the data.
- [Link to Lecture 3](#)

Module 9: Categorical Data & Chi-Square Tests

Module Assignment

The course outline explicitly lists chi-square and two-by-two tables, but the supplied PDF does not contain a dedicated full section on OR/RR. The assignment therefore follows the requested syllabus.

1. Construct a 2×2 table for smoking status versus lung disease using a simulated sample.
2. Calculate the relevant row, column and overall proportions.
3. State H0 and H1 for a chi-square test of independence.
4. Perform or obtain a chi-square test and interpret the p-value.
5. Calculate and interpret an odds ratio and relative risk.
6. State one limitation of interpreting association as causation.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.